Referee-style report on
Matrix Theory Reloaded: A BPS Road to Holography
(Blair, Lahnsteiner, Obers, Yan)

Summary. The paper treats the BPS decoupling limit that yields BFSS matrix theory as a limit of type II string theory as a whole, not only of the D0-brane worldvolume. The result is a "matrix p-brane theory" (MpT) with a codimension-(p+1) non-Lorentzian (Newton–Cartan-type) target space. The paper then does three things. (i) It shows that the same limit, applied asymptotically to the Dp-brane supergravity solution, produces the Maldacena/IMSY near-horizon limit. Stacking a second limit on an "off-aligned" Dq solution gives non-Lorentzian bulk geometries, and these reproduce the recent non-Lorentzian holography proposals of Lambert *et al.* (ii) Uplifting to M-theory, where each BPS limit corresponds to a DLCQ, the paper conjectures that holography is generally a DLCQ^n/DLCQ^m correspondence with $m > n$. (iii) It interprets undoing the limit as a p-brane generalisation of $T\bar{T}$, and derives classical flow equations that take free scalars to Dirac–Nambu–Goto in any dimension, and SYM to DBI exactly for $p = 1, 2$ (for $p \geq 3$, only perturbatively).

1. Principal achievement

The main achievement is conceptual and organisational. The paper shows that *one operation*, the asymptotic BPS (MpT) decoupling limit with a critical RR potential, simultaneously (a) reduces the open-string sector to matrix theory/SYM, and (b) induces the near-horizon limit in the closed-string (supergravity) sector, and that iterating it gives a systematic generator of holographic pairs. As a result, the matrix-theory and AdS/CFT decoupling limits are no longer separate prescriptions. They are two faces of a single limit, organised by duality orbits. The scheme makes some things explicit that are usually left implicit:

- the flat "boundary" geometry seen by SYM is intrinsically non-Lorentzian;
- the near-horizon solution asymptotes back to that geometry, with $(r/\ell)^{(7-p)/2}$ playing the role of ω via the emergent dilatation symmetry;
- the condition $q = p + 4$ (or smearing/rescaling otherwise) for an MpT limit of a Dq solution maps neatly onto the web of non-Lorentzian dualities found bottom-up by Lambert *et al.*

The most concrete and checkable new result is the closed-form flow equation

$$\partial_t \mathcal{L} = \tfrac{1}{2t^2}\Big\{ \mathrm{tr}(\mathbb{1} - t\mathcal{T}) - (p-1)\big[\det(\mathbb{1} - t\mathcal{T})\big]^{1/(p-1)} - 2 \Big\},$$

which deforms D free scalars in p+1 dimensions into the static-gauge Nambu–Goto p-brane. I checked it numerically for $p = 2, 3, 4, 5$ with several scalars and a generic constant embedding, and it holds. It reduces to $T\bar{T}$ for $p = 1$ and to the Gross–Kruthoff–Rolph–Shaghoulian deformation for $p = 0$. It also makes clear why the multi-scalar case differs from the single-scalar "root" flows used elsewhere. The $p = 1$ flow with a gauge field also explains why ordinary $T\bar{T}$ applied to 2d Maxwell does not end at DBI. This is a useful result for the field-theory community in its own right.

A caveat on novelty: for $n = 0$ the paper itself shows that its limit equals the Maldacena limit up to a reparametrisation of α'. Much of Sections 3–4 is therefore a (clarifying) reformulation. The genuinely new content is the systematics of the iterated limits and the flow equations.

2. Principal drawback

The central holographic claims are *kinematic*, and the paper tests none of them dynamically. The DLCQ^n/DLCQ^m conjecture is a classification scheme: it counts BPS limits on each side of a known or proposed duality. Not one observable is computed on both sides of a new non-Lorentzian pair, whether a spectrum, a correlator, a symmetry algebra or a thermodynamic quantity. Several foundational points are left open, as the authors partly acknowledge:

- **The bulk geometries are not shown to be solutions.** The non-Lorentzian near-horizon backgrounds are not checked against the MpT supergravity equations of motion or the torsional constraints. MpT supergravity is not even formulated. Without this, "bulk dual" is a statement about a limit of a solution, not about a solution of the limiting theory.
- **Smearing and N-rescaling.** Many entries of the landscape (beyond the $q = p + 4$ case) need smearing or a formal rescaling of N (effectively $N < 1$ at finite ω). The physical meaning of these steps is unclear, so it is not clear which entries are genuine dualities.
- **Heuristic DLCQ counting beyond $n = 1$.** The identification beyond $n = 1$ is suggestive rather than derived: for example, Spin Matrix Theory as DLCQ^2/DLCQ^3 rests on an R-charge limit that "may be associated" with a third DLCQ. There is also no argument that the bulk and boundary limits commute in general.
- **Question 3 is not answered intrinsically.** The $T\bar{T}$ answer to "how is the bulk generated intrinsically?" promotes $\omega \to \not\leqslant(r)$ by hand and fixes $\not\leqslant(r)$ by matching to the known solution. No equation determines $\not\leqslant$ from the deformation alone.
- **The p-brane $T\bar{T}$ flows are classical, bosonic and abelian.** They are derived by construction from the known endpoint and depend explicitly on t. Nothing is said about whether they keep any of the quantum solvability that makes 2d $T\bar{T}$ special, such as factorisation or an exact spectrum.

In short, the framework is elegant and self-consistent at the level of limits, but its holographic predictive power is so far untested.

3. Most worthwhile follow-up (feasibility × impact)

Proposal: turn one DLCQ^1/DLCQ^2 example into a tested duality by working intrinsically in M0T supergravity, using the M0T limit of the D4-brane near-horizon geometry (dilatation-equivalent to an M0T limit of AdS$_6 \times S^4$) as a calibrated test case.

(a) *Build the bulk theory.* Obtain the bosonic M0T supergravity equations, with their torsional constraints, as the $\omega \to \infty$ limit of IIA supergravity under the prescription the paper gives explicitly. The membrane analogue already exists (Bergshoeff *et al.*; Lambert *et al.*), so this is well-trodden technically. Then verify that the M0T near-horizon D4 background (Section 3.2 of the paper, with $p = 0$, $q = 4$) solves the equations. This settles the most basic open issue in Section 2.

(b) *Match symmetries.* Compute the (super)isometry algebra of the background as an Inönü–Wigner-type contraction, using the paper's M0T-algebra techniques. Compare it with the spacetime symmetry of the non-relativistic 5d SYM / null-reduced $(2, 0)$ theory of Lambert *et al.*

(c) *Match a protected spectrum.* Solve linearised fluctuations on the non-Lorentzian background at fixed D0 charge N. Compare with BPS states of quantum mechanics on instanton moduli space, which are computable via an index.

Why this example. The D4/M0T case is the one entry that needs no smearing or N-rescaling. It also uplifts to $\mathrm{AdS}_7 \times S^4$ with a null circle, so every bulk answer can be cross-checked independently against known AdS_7 results restricted to fixed null momentum (and against the DLCQ description of the $(2,0)$ theory). This makes the project low-risk. Its value is that it tests the *framework itself*: the MpT equations, the torsional constraints and the dilatation symmetry, against a known answer.

Extension. Once calibrated, the same machinery can be applied directly to entries with no simple uplift, i.e. the smeared and N-rescaled backgrounds. There it would settle whether those backgrounds define genuine dualities. This would be the first non-trivial test of the DLCQ^n/DLCQ^{n+1} conjecture beyond $n = 0$, and it would provide the MpT supergravity toolkit that the rest of the programme (finite temperature, isometries, further DLCQs) needs.

Runner-up. A cheaper but more speculative project: study the $p = 2$ flow equation, $\partial_t \mathcal{L} = \frac{1}{4}[\mathrm{tr}\,\mathcal{T}^2 - (\mathrm{tr}\,\mathcal{T})^2] + \frac{t}{2}\det \mathcal{T}$, at the quantum level. Two questions are natural: does it admit any exact spectral formula on T^2 for BPS states, and can it be applied to ABJM? This is higher-risk, but would have broad field-theory impact if it worked.